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Effects of velocity on evaporation rate of nanoscale water droplets

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ABSTRACT

The evaporation of nanoscale water droplets exhibits significant deviations from continuum-based predictions, posing challenges for accurate modeling. Yet, the accurate prediction of evaporation process can be crucial to many natural and engineering applications. This study employs molecular dynamics (MD) simulations to investigate the impact of air velocity on the evaporation dynamics of nano water droplets. Previous research demonstrated that evaporation rates of nanodroplets increase with both temperature and initial droplet size. However, these studies were mostly limited to stationary droplets. The work presented in this paper reveals a counter-intuitive decrease in evaporation rate with increasing air velocity. This study attributes and proves (through MD data analysis) that this phenomenon is due to the compressive forces exerted by moving air molecules, which reduce the kinetic energy of surface water molecules, thereby inhibiting evaporation. The findings underscore the importance of considering nanoscale effects in evaporation models and suggest potential applications in fields such as disease transmission, coating technology, and nanoscale drug delivery.

1. Introduction

It has been shown that evaporation of droplets at the nanoscale deviates significantly from continuum observations, limiting the use of continuum-based laws to predict the behavior of nanodroplets evaporation [1]. Yet, evaporation of droplets is a fundamental phenomenon, and capturing and predicting this behavior is critical for many natural and engineering applications [1–5]. Evaporation of nano-droplets, however, is challenging to observe by experimental means today due to limitations in technology to observe and quantify phenomenon at nanoscale [6]. While there are some recent experimental reports on sessile nanodroplets [6], there are no such experiments known to the authors conducted to study evaporation of moving nanodroplets.

Molecular dynamics (MD) simulations provide an alternative means to observe and quantify nanodroplet evaporations. MD is a simulation technique that models individual atoms and their interactions with other atoms [7,8]. Based on the interaction dynamics, forces on each atom at every time step are computed and the velocities and positions are subsequently updated by numerical integration of Newton's second law [9]. Due to the lack of simplifying assumptions, MD is widely used to study and observe molecular phenomena, which are currently still very challenging and costly in physical experiment [10]. Where experimental observation was possible, the predictions from MD simulations can provide very good agreement [11–13].

In this work, MD is used to investigate the effect of air velocity on the evaporation dynamics of nano water droplets. The effects of temperature and initial droplet size on the evaporation rate of suspended nano water droplets were previously investigated [1]. Evaporation rate was found to increase with both temperature and droplet size for a nanoscale water droplet. It is not surprising that the evaporation rate was found to be increasing with increasing temperature, since the kinetic energy of atoms are higher at higher temperatures, resulting in high probability of water molecules breaking away from the bulk droplet at the interface. However, counter-intuitively, nano water droplet's evaporation rate was found to also increase with increasing initial droplet size. It was postulated that at the nanoscale, the increased surface area at the interface which results in more molecules having a smaller number of hydrogen bonds dominates, resulting in an increased rate of evaporation observed as nanodroplet size increased. These observations were made with droplets that are stationary. Other related works also mainly dealt with stationary or sessile nanodroplets. For suspended droplets, Hou et al. studied the evaporation dynamics of metal nitrate containing suspended nanodroplets using MD [14]. MD revealed that in metal nitrate containing nanodroplets, as water molecules leave the droplets, the metal nitrate tends to precipitate on the droplet surface forming a solver-core-solute-shell structure. In other separate studies, Wang et al. studied the

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evaporation of suspended alkane/alcohol droplets [15], while Liu et al. studied the impact of acoustic field on suspended nanodroplets evaporation [16]. Pestana and Head-Gordon studied the evaporative behavior of sessile nanodroplets and observed a significant difference with their evaporative behavior when compared to macroscopic droplets [17]. They observed deviations from continuum d^2 law predictions, and additionally observed different evaporation kinetics with varying hydrophobicity of the nanodroplets. Li et al. similarly used MD to study sessile nano-droplets but studied instead the effect of roughness of the substrate on the evaporation rate [18]. In their work the difference in behavior of sessile droplets at nano and macro scales is again highlighted.

As far as the authors are aware, there is currently no systematic MD study carried out to understand how air velocity can affect water nanodroplets' evaporation. Based on macroscope scale observations, the evaporation rate is expected to increase with increasing air velocity [19,20]. However, several MD studies have observed significant deviation from macroscopic observations when continuum law does not apply [17,18]. In our previous work [1], similar deviation from continuum law was also observed for suspended nanodroplets. It was found that nanodroplet evaporation occurs at a much slower rate than predicted by continuum models at varying temperatures. Therefore, absolute care must be taken when using continuum models, such as the d^2 law [21], for nanodroplets. In this contribution, we report on the MD study of evaporation of moving droplets at different air velocities. It was found that evaporation rate decreases with air velocity at the nanoscale, counter-intuitively. By investigating the radial pair distribution between water molecules in the droplet, studying the pressure distribution around the nanodroplet, and calculating the kinetic energy of the surface water atoms, it can be concluded that the evaporation rate decreases as the water molecules are being compressed closer together. This study elucidates the counter intuitive observation of water nanodroplets evaporation rate with increasing velocity.

At the same time, while MD simulation can be a reliable tool to predict evaporation rate at the nanoscale, its high computational power requirement limits its range of applicability. Therefore, it is postulated that the use of data-driven methods to supplement MD simulations can be a plausible way to allow scientists and engineers to easily estimate the evaporation rate of nano water droplets as needed. This is especially important as there are currently limitations in the use of experimental systems to observe evaporation at such small scale. The pre-requisite is to build a sufficiently large database that can allow the data driven model to be built. In our previous work, we built a dataset that can be used to predict evaporation rate of nano-scale water droplets with varying initial droplet size, and at varying temperature. In this work, we advanced the dataset to estimate the impact of air velocity on the evaporation rate. This dataset is hosted at https://github.com/elisaaym/Water_nanodroplets_evaporation.

2. Molecular dynamics simulation methdology

MD simulations are performed using the large-scale atomic/molecular massively parallel simulator (LAMMPS) [22] and results are post-processed using the Open Visualization Tool [23]. The TIP4P water model with a long-range coulombic solver, with the particle-particle particle-mesh (PPPM) solver [24], is used to model the behavior of the water molecules in the present simulation [25]. The TIP4P water model was chosen due to its ability to model the hydrogen bonding network and its agreement with experimental data [26]. The simulation domain is illustrated in Fig. 1. Water droplet of diameter D = 4 nm is fixed right in the middle of a periodic domain that has 5 nm distance from the edge of the droplet to the end of the domain. To model the air around the domain, sufficient O_2 and N_2 molecules are added to surround the water droplet to model the density of air at standard atmosphere. The harmonic bond potential defined in [27,28] is used to model the intra molecular interactions within O_2 and N_2 molecules. Intermolecular

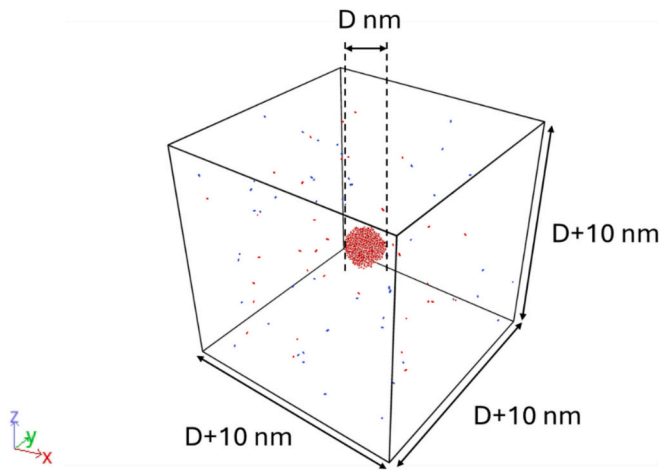


Fig. 1. Illustration of the simulation domain adopted in this molecular dynamics study.

interactions between O_2 and N_2 and water molecules are based on the Lorentz-Berthelot mixing rule, with the intermolecular Lennard-Jones potential of O_2 and N_2 adopted from [27,28].

To simulate moving droplets, due to computational limitations, air molecules are prescribed to move in the x direction instead. This is achieved by applying force F on all air molecules until the air molecules eventually reach an equilibrium velocity. This effect is illustrated in Fig. 2 for the different forces considered. Production run begins only after equilibrium velocity is reached. As seen in Fig. 2(b), air velocities ranging from 0 to 1200 m/s are considered in this work. This range is chosen to effectively capture the change in evaporation rate observed with varying airspeed, as would be discussed in Section 3.

All simulations begin with an equilibration at isothermal-isobaric conditions (NPT) [29] with a timestep of 0.5 fs for 100,000 timesteps. During the NPT equilibration, the temperature of the domain was allowed to increase from 293 to 303 K. This is then followed by a further equilibration at NVT conditions, at constant temperature of 303 K at a timestep of 1 fs for 100,000 timesteps. Production runs at 303 K then commences, with a simulated runtime of 30 ns. The resulting evaporation dynamics reported herein occur only during the production runs. The production run is performed under canonical (NVT) ensemble. The rationale for isothermal assumption is the assumed scenario of having a small nanodroplet surrounded by a large body of air, which therefore acts as the thermal reservoir. The small nanodroplet is assumed to be the body moving through air, and therefore any heating near the nanodroplet will be dissipated quickly with the large body of air surrounding the nanodroplet. If one would like to consider a rise in temperature of ambient air due to heating up of the air from the high velocity, one can no longer assume the canonical ensemble. In this case, the evaporation mechanics should be considered using the micro-canonical ensemble (NVE) condition. Supplementary Information Section S1 reports a separate simulation run under the NVE condition, where the ambient gas is heated up due to the increase velocity, causing evaporation to occur quickly. This is not the case considered in this paper, and the additional simulation run is just performed for illustrative purposes.

Additionally, in the simulations, water molecules are removed after they leave the droplet surface and travel past 10 Å from the middle of the domain, as illustrated in Fig. 3. This is to ensure that the relative humidity (RH) of the surrounding air remains close to the desired RH, in this case close to 0 %. Due to the small size of the domain and the periodic boundary conditions imposed, the presence of one water molecule is sufficient for the air to significantly alter the RH of the surrounding air. Hence, it is essential to remove water molecules from the simulation domain periodically as they evaporate. Similar handling of evaporated water vapor molecules are used in other molecular dynamics study of

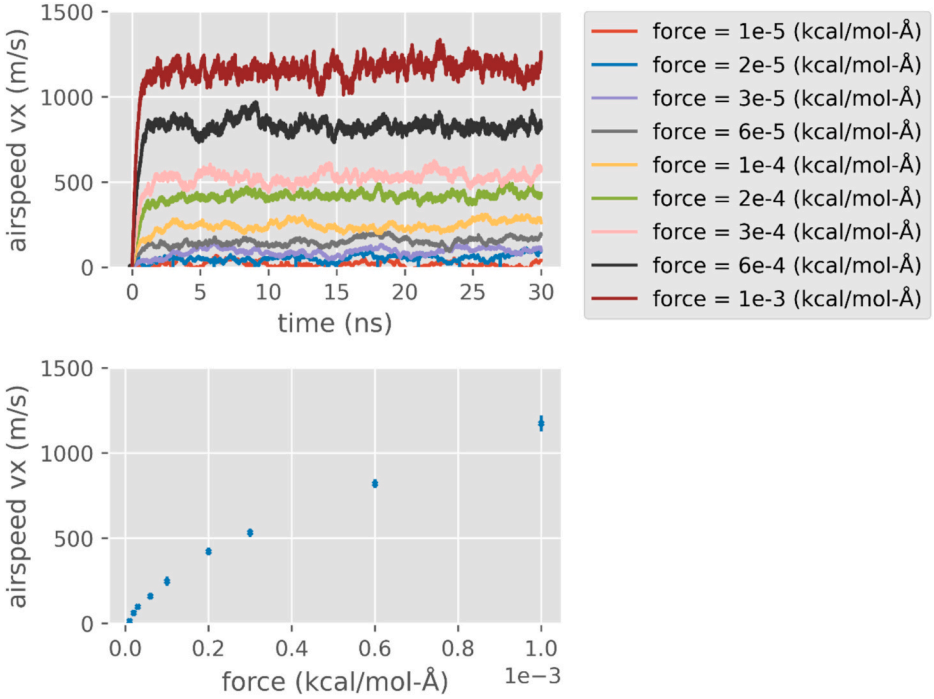


Fig. 2. Velocity of the air molecules when different forces are applied. Note that the unit of force presented is kcal/mol-Å, which is a unit that is relevant at the nanoscale.

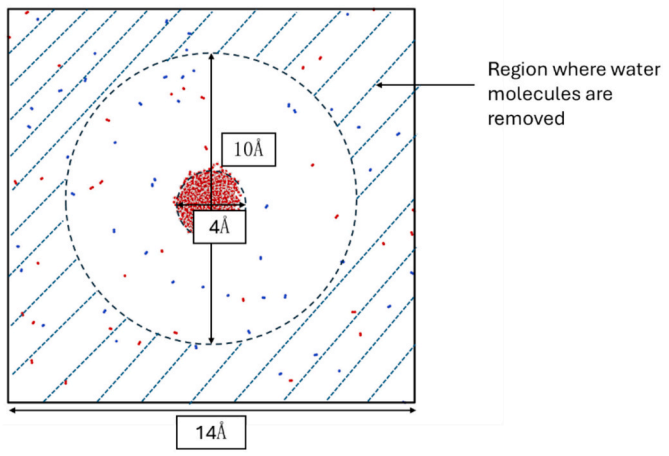


Fig. 3. Illustration of how water molecules are removed in the simulation when they leave the droplet.

water evaporation [30,31]. This is carried out in LAMMPS via the fix evaporate command after the water molecules leave the surface of the water droplet. The number of water molecules remaining in the droplet over time is monitored.

The evaporation rate for this droplet can then be estimated from the gradient of the linear regression line fitted, as illustrated in Fig. 4. The measurement of evaporation rate via monitoring the change in number of remaining water molecules is the most direct way to measure evaporation rate in molecular dynamics [14,31–33], and hence, is adopted in this study. Each simulation is carried out once, however, a check was done to ensure the consistency of data by repeating the simulation thrice for the case when force = 0 and force = 1e-3 kcal/mol-Å, for different initial configurations. Details are presented in Supplementary Information Section S2. As can be seen from Fig. S2.1, the runs are very repeatable.

3. Results and discussion

With the analysis methodology outlined in Section 2, the evaporation rate of a 4 nm droplet at 303 K, with varying airspeeds is computed for the airspeed range considered in this work, and the results are presented in Fig. 5. Results reveal a decrease in evaporation rate as airspeed increases, which is counter intuitive from what one would expect from continuum observations. At low air speeds from 0 to about 150 m/s, the evaporation rate remains almost constant at approximately 9 molecules per ns. There is a slight increase in evaporation rate observed as airspeed increases from 0 to about 10 m/s, where evaporation rate increases from about 8.5 to 9 molecules per second, but thereafter the evaporation rate remains constant at about 9 molecules per second until 100 m/s. After 100 m/s, there is a sharp drop in the evaporation rate from 9 molecules/ns, and approaches almost 0 at 800 m/s. From 800 to 1200 m/s, the evaporation rate of the nanodroplet is observed to be almost negligible. While the lower velocity ranges below 100 m/s are seen in most typical applications for nanodroplets today, the higher velocity ranges in the order of 100 m/s to 1200 m/s may be relevant with the new discovery of ultrafast propulsion of nanodroplets on different patterned surfaces or nanotubes [34–37], or in specialized applications where ultrafast nanodroplets transport are required.

Supplementary Video 1 showcases the difference in droplet dynamics at airspeeds of 0 m/s, at 100 m/s and at 1200 m/s. At an airspeed of 0 m/s, the droplet is visibly less compressed and the water molecules at the surface are seen to stretch more from the bulk droplet. At 1200 m/s, the droplet is much more compressed and rigid, and the water molecules at the surface do not stretch much from the bulk droplet. From our previous analysis [1], evaporation of a stationary water nanodroplet is governed by two main physical forces. First, the attractive intermolecular force between water molecules tends to hold the droplets together, preventing evaporation. Second, the kinetic forces of each water molecule and their interactions with the surrounding air molecules tend to pull water molecules away from the water droplet. At the water-air interface, there is a lower intermolecular attractive force due to the reduced number of hydrogen bonds, and that results in the kinetic forces having a higher probability of pulling the water molecule away,

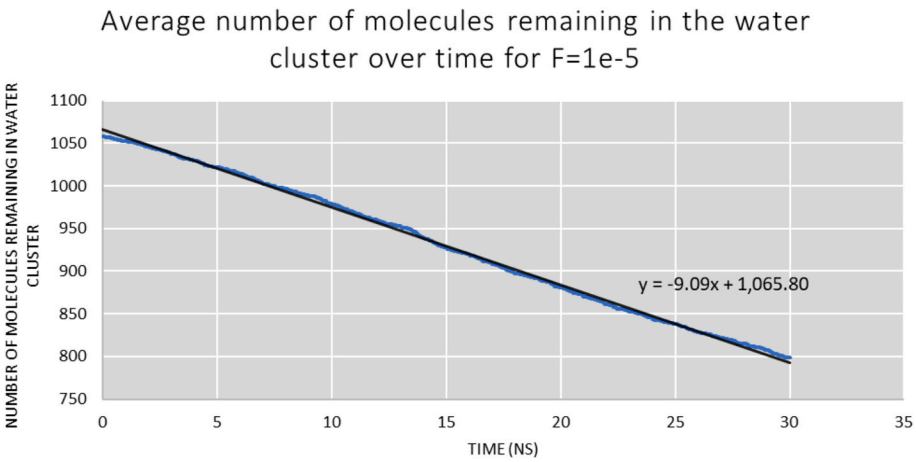


Fig. 4. Illustration of how the rate of evaporation is obtained from monitoring the number of molecules remaining in the water droplet over time (example shown is for F = 1e-5). The blue line is the observed variation, and black is the fitted line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

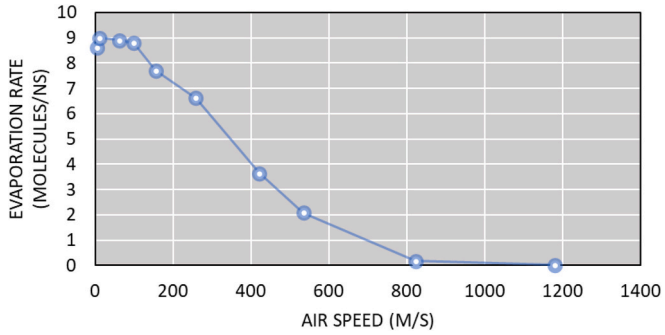


Fig. 5. Decreasing evaporation rate observed with increase in airspeed for 4 nm sized nanodroplet at 303 K.

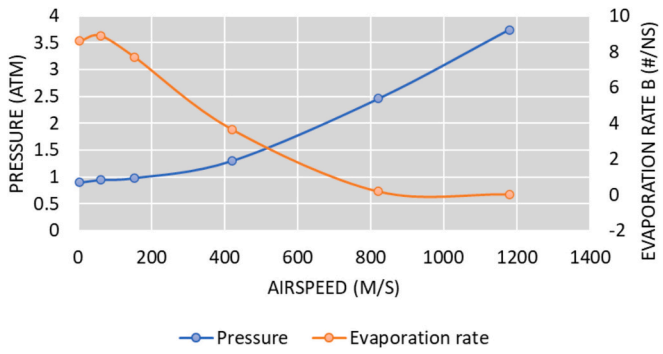


Fig. 6. Domain pressure with varying airspeed.

resulting in evaporation. At the nanoscale, it was observed that the larger the droplet, the larger the surface area of the water-air interface, and hence, larger evaporation rate is observed. Now with the addition of air velocity, a third force is at play. That is the effective pressure force acting on the droplet due to movement of the air molecules with respect to the droplet. Due to the movement of air molecules around the droplet, there is a net compressive force acting on the droplet due to Bernoulli's effect [38]. This pressure force tends to hold the droplet together. The increase in air velocity around the droplets tends to increase the kinetic forces that tend to pull the water away from the droplet, but at the same time, causes an increase in compressive force that tends to prevent the water molecule from escaping.

To quantify these forces at play, we performed further investigation as summarized in the sub sections below.

3.1. Domain pressure

We first plotted the domain pressure with respect to increasing air velocity. The domain pressure was averaged at the last 3 ns of the total simulation time (30 ns). The resulting domain pressure with respect to the different airspeeds investigated is shown in Fig. 6. On the same figure, the variation of the evaporation rate with airspeed (as presented in Fig. 5) is also plotted for clarity. With the increase of air speed beyond 100 m/s, there is a sharp rise in domain pressure, coinciding with the sharp drop in evaporation rate. Domain spressure continues to increase to about 3.5 atm at about 1200 m/s of airspeed. However, the evaporation rate stops decreasing fairly abruptly at about 800 m/s as it is already at a value close to 0. The increase in domain pressure is expected to compress the water droplet, resulting in the reduced evaporation. To support this hypothesis, we additionally looked at the Oxygen-Oxygen (O—O) radial distribution function (RDF) and the radial density profile (RDP), described in the Section 3.2 and 3.3 respectively.

3.2. Oxygen-Oxygen (O—O) radial distribution function (RDF)

The Oxygen-Oxygen (O—O) radial distribution function (RDF) is illustrated in Fig. 7. The O—O radial distribution function shows the probability of finding an oxygen atom a certain distance away from another oxygen atom [39]. Fig. 7(a) shows the O—O radial distribution function of the simulation domain. The first peak at 1.25 Å corresponds to the O—O distance between two oxygen molecules, and the second peak at about 2.72 to 2.78 Å corresponds to the median O—O distance between two water molecules. This is in good agreement with reported experimental or simulated data [40]. Zooming into this peak (depicted in the purple dotted box of Fig. 7(a), one can see that with increasing airspeed, the O—O RDF of water shifts from 2.78 Å to 2.72 Å. This is plotted in Fig. 7(b) for clarity.

3.3. Radial density profile (RDP) and droplet core density

The RDP is measured by considering concentric spheres around the droplet, each of width 2Å, as illustrated in Fig. 8(a). The number of water molecules within each concentric sphere is then counted. By normalizing this quantity with the volume of concentric sphere, we obtain the RDP, which is the number density of the droplet at different radial distance from the center. The RDP is obtained at regular intervals

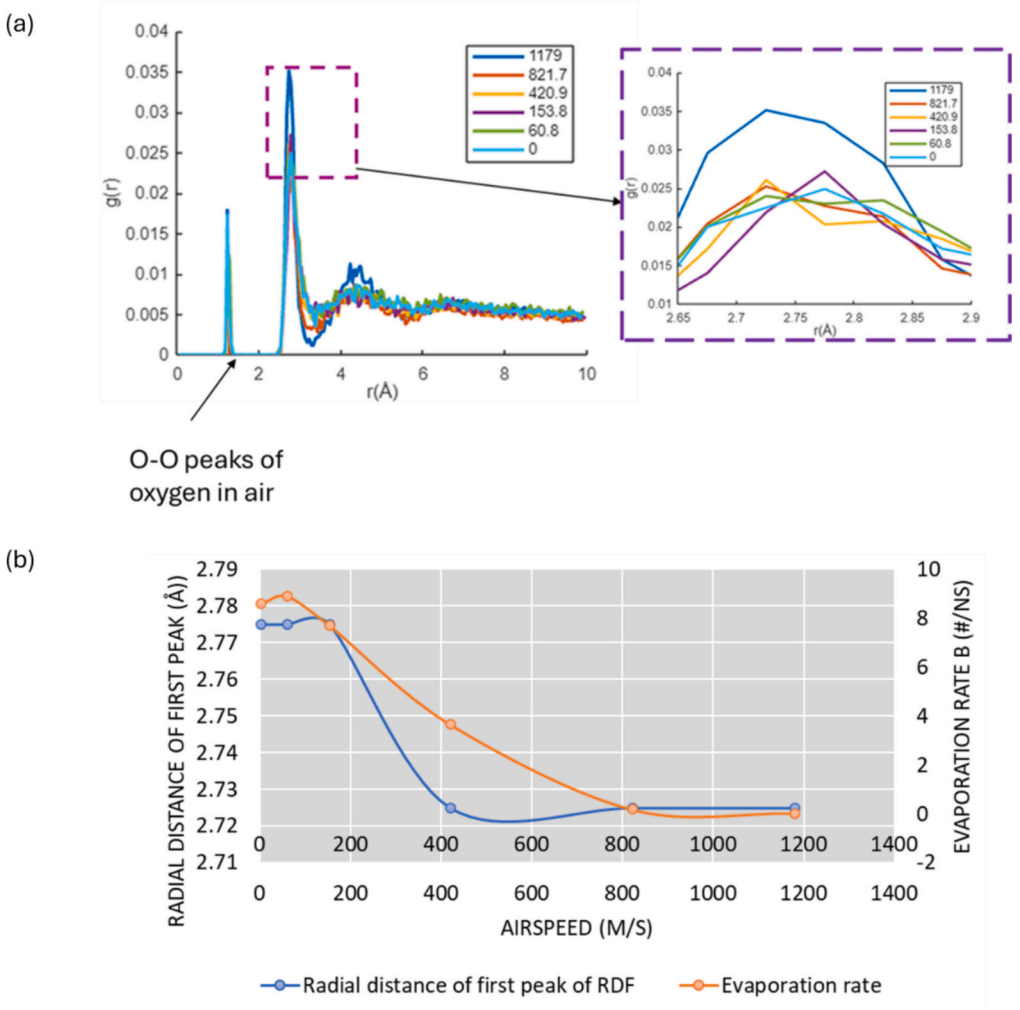


Fig. 7. (a) O—O Radial distribution function (RDF) profiles at different airspeed (b) Plot to show the variation of first peak of O—O RDF with airspeed.

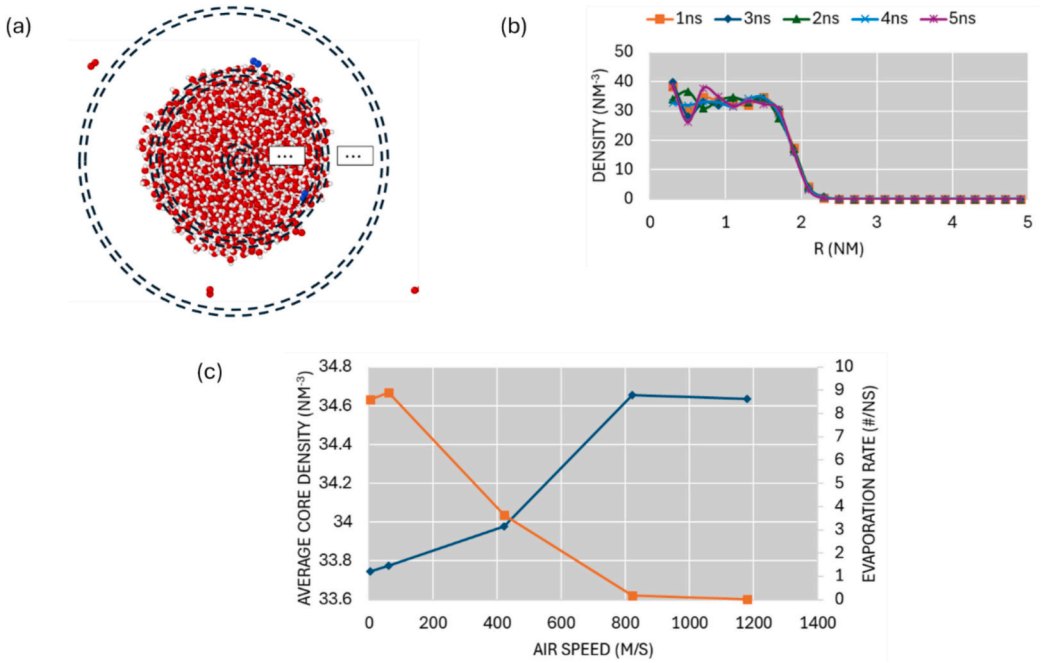


Fig. 8. (a) Calculation of radial density profile (b) RDP (c) variation of core density of the droplet with airspeed.

from 1 to 5 ns in the production run to ensure that the RDP is no longer changing with time. This is illustrated in Fig. 8(b) for F = 0 (airspeed = 0 m/s). The core density of the water droplet can then be obtained by considering the constant density within the core of the droplet, which occurs below 1.5 nm radius as shown in Fig. 8(b). The average core density with respect to other air speeds are presented in Fig. 8(c). As can be seen, the average core density increases with air speed, consistent with the increase in domain pressure measured, as well as the reduction in O—O RDF. These analysis supports the described mechanism that the motion of air molecules generates compressive forces through the Bernoulli effect, which inhibits evaporation.

3.4. Surface tension

To further quantify the compressive force on surface water molecules, we further computed the surface tension. We follow the method from Leong & Wang (Leong & Wang, 2018). Using Young-Laplace equation, surface tension σ can be obtained from isotropic internal pressure as given in Eq. 1.

$$\Delta P = \frac{2\sigma}{R} \tag{1}$$

Where ΔP is the pressure difference between the bulk-like center and the vapor, and R is the radius of the droplet. To calculate ΔP, we correlate density with pressure by running a series of bulk water simulations with 8000 water molecules. The bulk water simulations are run with isothermal-isobaric (NPT) conditions to control the pressure, and temperature. The density can then be measured to obtain a pressure-density relation for the pressure range we are interested in. The temperature was set at 303 K, and the pressure varied from 1 to 1000 ATM. The resulting pressure-density relation is shown in Fig. 9. The same trend of increase liquid pressure with density as reported in (Leong & Wang, 2018) is observed. Following (Leong & Wang, 2018), a quadratic equation is fitted to the pressure-density curve to obtain Eq. 2, which allow correlation of droplet average core density with the pressure difference between the droplet internal core and the ambient air ΔP.

$$\Delta P = 44.4\rho^2 - 2300.9\rho + 27804 \tag{2}$$

Using Eq. 1 and 2, we obtain the surface tension of the droplet with increasing airspeed, presented in Fig. 9(b). At an airspeed of 0 m/s, the measured surface tension is about 70 mN/m, close to the theoretical surface tension of water. With an increase in airspeed, the surface tension is seen to increase, which further confirms the compressive effect observed, which causes higher intermolecular forces of attraction between the surface water molecules. This is aligned with the observations of increased core density, and decrease O—O RDF observed in Section 3.2 and 3.3.

3.5. Surface kinetic energy

Lastly, we also quantified the kinetic energy of the layer of water molecules closer to the surface. This is carried out by computing the average kinetic energy of water droplets between the radial distance 2.5 and 4 nm from the centre of the droplet, as depicted in Fig. 10(a). The trend observed is similar to that observed on the variance of domain pressure with airspeed depicted in Fig. 6. It is therefore hypothesized that the kinetic energy of the outer layer of the droplet reduces due to increase domain pressure with increasing air speed. The compressive force reduces the kinetic energy of the droplet, which then directly accounts for the lower rate of evaporation, since there is less kinetic energy to overcome the forces of attraction to enable evaporation.

3.6. Evaporation rate variation with airspeed at elevated temperature

To further validate the above findings, a set of MD simulations is run at an elevated temperature of 343 K at different airspeed. Since droplets evaporate faster at 343 K compared to 303 K, the simulation run time was capped at 10 ns at 343 K. The results are shown in Fig. 11. Similar trends are observed as compared to the runs conducted at 303 K, where the evaporation rate decreases rapidly when airspeed is more than 100 m/s, but moderates to almost 0 at 800 m/s. It can also be observed that as compared to Fig. 5 for the case of 303 K, the evaporation rates at lower airspeeds are much higher when temperature increased to 343 K. While the evaporation rate starts at about 9 molecules/ns at 303 K when airspeed is 0, at 343 K, the evaporation rate is about 40 molecules/ns. Therefore, it is clear that the observed behavior of the droplets evaporation rate decreasing with airspeed is not dependent on temperature of droplets.

4. Conclusion

This work highlights that at the nanoscale, water droplet's evaporation rate actually decrease with increasing airspeed. This is rather counter-intuitive and is contrary from that observed in the macroscale [19,20]. At lower airspeeds below 100 m/s, evaporation rate is maintained at about 9 molecules per ns. However, as airspeed increases from 100 m/s to 800 m/s, the evaporation rate dropped rapidly to almost 0 molecules per ns, and further increasing the airspeed to 1200 m/s does not lead to any increase in the evaporation rate. The same phenomenon is observed at both 303 K and 343 K. In this contribution, the reason for this phenomenon is investigated in detail by considering the domain pressure, O—O RDF, droplet core density, surface tension and surface kinetic energy of the droplet. The observation reported here in is summarized in Table 1.

A fundamental understanding of the evaporation phenomenon at nanoscale is important for applications such as dispersion prediction for disease or toxic chemical transmission [41], or applications such as coating technology, microelectronics cooling or inkjet printing [42–44]. The negligible evaporation observed at high speed can also help to shape developments of very fast jets with nanodroplets, or nanodroplets transport or nanoscale drug delivery. This is especially so as researchers and scientists are discovering ultrafast propulsion of transport of

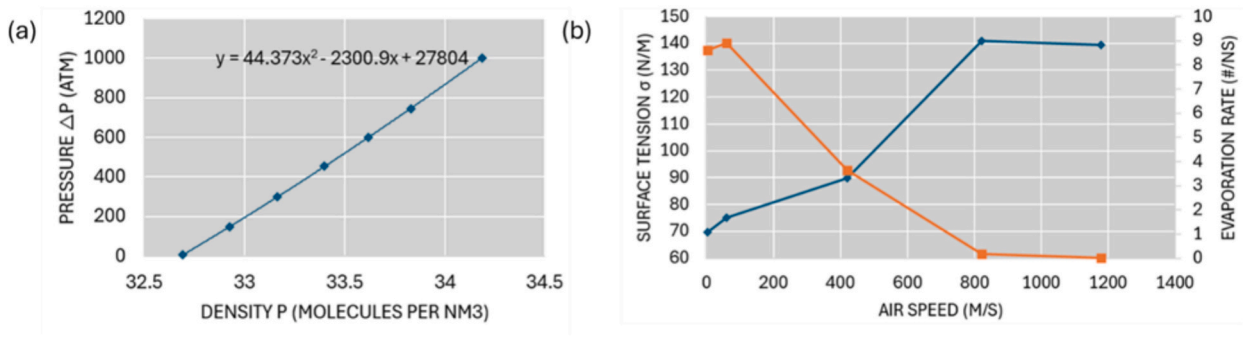


Fig. 9. (a) Pressure-density relation obtained at 303 K (b) computed surface tension with respect to airspeed.

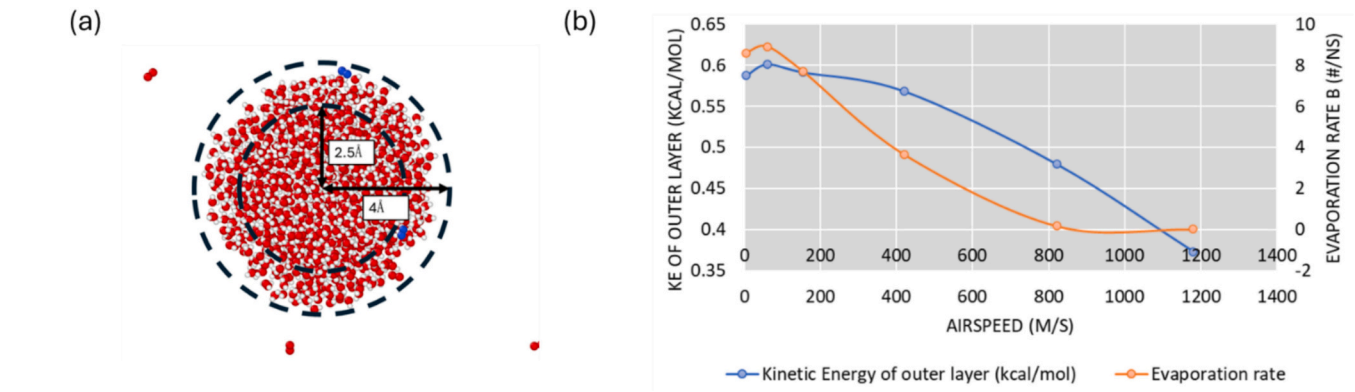


Fig. 10. (a) Depiction of outer layer of water molecules close to the surface of the water droplet where kinetic energy is computed. (b) Kinetic energy of the outer layer of water against airspeed.

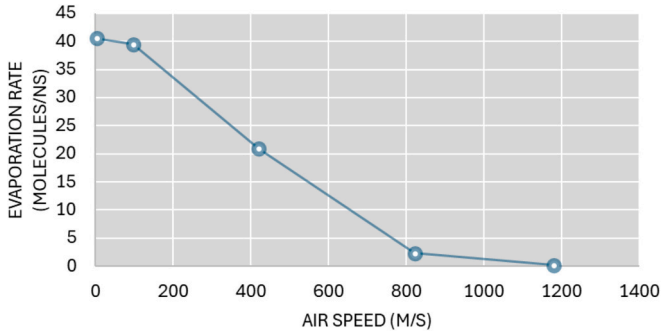


Fig. 11. Decreasing evaporation rate observed with increase in airspeed for 4 nm sized nanodroplet at 343 K.

Table 1
Summary of the observations from this investigation that explains the reduction of evaporation rate with increasing airspeed.

Measurable quantity	Observation
Domain Pressure	Increases with airspeed
O-O RDF	Decreases with airspeed
Droplet core density	Increases with airspeed
Surface tension	Increases with airspeed
Kinetic Energy at the surface of the droplet	Decreases with airspeed
Evaporation rate	Decreases with airspeed

nanodroplets on different patterned surfaces or nanotubes [34–37], where velocities of droplets are in the order of 100 m/s.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.icheatmasstransfer.2025.108851.

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CRediT authorship contribution statement

Elisa Y.M. Ang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Peng Cheng Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **William Toh:** Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Teng Yong Ng:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Peng Cheng Wang reports financial support was provided by Government of Singapore Ministry of Education. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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